TrainerBot: two-function pipeline

Sensors → software → results. Interface: choose the mode, Start / Stop.

1 Band exercise: measure arm angles

Camera
Films the arm

→

MediaPipe + OpenCV
Gets image from camera and finds shoulder, elbow & wrist

→

Python
Calculates the elbow angle

→

Angle results
Show and save angles over time

The robot stays still during this exercise.

2 Touch game: alternate between two sides

Top touch sensor
Detects a press and release

→

Python + ROS 2
Chooses the opposite side

→

Robot wheels
Move around the chair to the other side

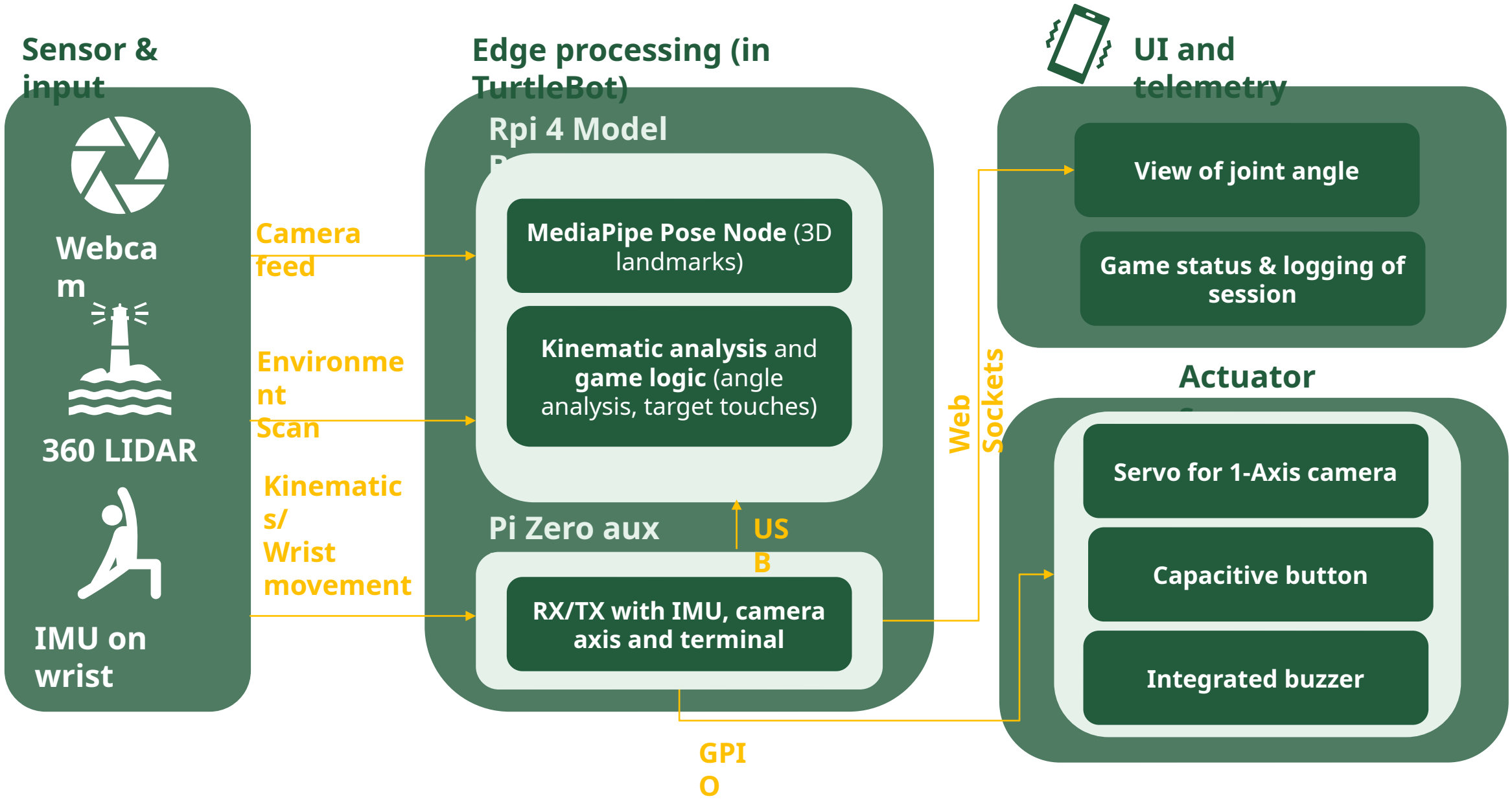
→

Stop and wait
Wait for another press; then repeat

Each press starts the next move. Stop ends the game.

Status: angle estimation tested on sample video. Saving angles, the interface and the touch game are planned.

Pipeline


Sensor & input
Webcam
360 LIDAR
IMU on wrist
Camera feed
Environment Scan
Kinematics/ Wrist movement
Edge processing (in TurtleBot)
Rpi 4 Model
MediaPipe Pose Node (3D landmarks)
Kinematic analysis and game logic (angle analysis, target touches)
Pi Zero aux
USB
RX/TX with IMU, camera axis and terminal
GPIO
Web Sockets
UI and telemetry
View of joint angle
Game status & logging of session
Actuator
Servo for 1-Axis camera
Capacitive button
Integrated buzzer